

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | **Context:** Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology is calibrated for news discourse: GPT-4o-judged fluency (1–5), content errors, lexical cohesion errors, and grammatical cohesion errors [Q54, Q55, Q180, Q191], plus d-chrF and d-BLEU [Q51, Q52] and human direct assessment on a 0–100 scale [Q195]. The benchmark explicitly defines lexical cohesion as 'vocabulary usage, incorrect or missing synonyms' [Q189] — narrative discourse flow, not legal terminological consistency. No rubric component evaluates whether named entities such as cooperative tier designations, land-tenure category labels, or subsidy scheme names are rendered identically across all clauses of a document — the critical coherence requirement for the deployment. Web research confirms terminology-consistency metrics exist (Semenov & Bojar WMT 2022; HHI) but have not been validated for Amharic [WEB-25, WEB-26]. Compounding this, the benchmark itself reports that GPT-4o-as-judge is unreliable for African language translation directions [Q67, Q85, Q99, Q219], leaving human DA as the residual signal — which had Krippendorff's alpha of only 0.46 for Amharic [Q71, Q211]. Given the user's MODERATE priority on OO and the partial match (d-chrF captures morphological richness [Q53], cohesion definitions touch on relevant phenomena), score is 2 rather than 1.

ID	Evidence
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q52	"Metrics were computed using SacreBLEU (Post, 2018) with bootstrap resampling (n = 1000) to report 95% confidence intervals." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q54	"We use GPT-4o as a judge to evaluate translation outputs, following recent work showing LLMs' effectiveness in assessing translation quality and analyzing errors (Wu et al., 2024; Sun et al., 2025)." (p.5)
Q55	"Following Sun et al. (2025), we assess each translated document's fluency, content errors (CE), and cohesion errors—specifically lexical (LE) and grammatical (GE) errors—using GPT-4o, with evaluation limited to a few model outputs due to cost constraints (Appendix B.6)." (p.5)
Q67	"These inconsistencies raise concerns about GPT-4o's reliability." (p.7)
Q71	"We obtained Krippendorff's alpha values of ≥ 0.40 , which are relatively low due to the fine granularity of the evaluation scale." (p.8)
Q85	"However, our DA and qualitative analysis found GPT-4o's judgments inconsistent for African languages, and sentence-by-sentence translation proved more effective for some languages." (p.9)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q180	"GPT-4o is prompted to rate the fluency of a document on a scale from 1 to 5, where 5 indicates high fluency and 1 represents low fluency." (p.22)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q191	"Fluency can only have values between 1 and 5. However, the other metrics, including CE, GE, and LE, do not have a specific range and can take on any value because they are counts." (p.24)

Q195	"Each annotator was assigned 80 documents to evaluate, tasked with marking as many error spans as possible and rating the overall quality on a scale from 0 to 100." (p.24)
Q211	"We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively." (p.26)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
WEB-25	Semenov & Bojar (WMT 2022) terminology-consistency metric exists but not validated for African languages (https://aclanthology.org/2022.wmt-1.41/)
WEB-26	HHI applied to legal corpora terminology consistency — not validated for Amharic (https://www.researchgate.net/publication/358368108_Measuring_Terminology_Consistency_in_Translated_Corpora_Implementation_of_the_Herfindahl-Hirshman_Index)

Strengths

- d-chrF is explicitly chosen as primary because it captures morphological richness of African languages [Q53]
- Output ontology engages with discourse-level phenomena (cohesion, fluency) [Q187, Q188] that have some overlap with cross-clause requirements of legal documents
- Benchmark transparently flags GPT-4o judging unreliability for African languages [Q67, Q85, Q99, Q219], enabling informed substitution

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Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q67	"These inconsistencies raise concerns about GPT-4o's reliability." (p.7)
Q85	"However, our DA and qualitative analysis found GPT-4o's judgments inconsistent for African languages, and sentence-by-sentence translation proved more effective for some languages." (p.9)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q187	"Cohesion refers to how different parts of a text are connected using language structures like grammar and vocabulary." (p.23)
Q188	"It ensures that sentences flow smoothly and the text makes sense as a whole." (p.23)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories (fluency 1-5, CE, LE, GE, d-chrF, d-BLEU, DA 0-100) are partially relevant — fluency and content-error scoring map to deployment quality concerns, but the categories are calibrated to news adequacy rather than legal-administrative precision [Q189, Q190].

OO-2: Missing categories specific to the deployment: (a) terminological consistency across clauses for proclamation-derived terms — no metric exists [WEB-25, WEB-26 confirm metric gap for Amharic]; (b) structural fidelity preservation for numbered clauses; (c) register-appropriateness scoring against Ethiopian proclamation register; (d) cross-reference accuracy for proclamation article citations.

OO-3: Categories assume news-discourse decision rules — 'lexical cohesion mistakes involve... overuse of certain words that disrupt the flow' [Q189] explicitly penalizes repetition, which is the opposite of what is needed for legal terminological consistency where identical repeat-rendering is required.

OO-4: Stakeholder-driven taxonomy redesign is needed: deployment scoring should add (1) terminology-consistency (e.g., HHI-based [WEB-26]) for proclamation terms; (2) register-appropriateness assessed by federal bureau translators [elicitation Q3]; (3) structural-fidelity scoring if format preservation is required.

OO-5: Issues harming structural validity: scoring rubric structurally misrepresents the construct of legal-administrative translation quality (treats consistent term repetition as a defect rather than a virtue [Q189]). Issues harming content validity: missing terminology-consistency, register, and structural categories. External validity at risk: GPT-4o judging shown unreliable for African languages [Q99, Q219].

ID	Evidence
Q3	"We conduct document-level translation benchmark experiments by evaluating the ability of neural machine translation (NMT) models and large language models (LLMs) to translate between English and these languages, at both the sentence and pseudo-document levels, the outputs being realigned to form complete documents for evaluation." (p.1)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q55	"Following Sun et al. (2025), we assess each translated document's fluency, content errors (CE), and cohesion errors—specifically lexical (LE) and grammatical (GE) errors—using GPT-4o, with evaluation limited to a few model outputs due to cost constraints (Appendix B.6)." (p.5)
Q67	"These inconsistencies raise concerns about GPT-4o's reliability." (p.7)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q190	"Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses." (p.23)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
WEB-25	Semenov & Bojar (WMT 2022) terminology-consistency metric exists but not validated for African languages (https://aclanthology.org/2022.wmt-1.41/)
WEB-26	HHI applied to legal corpora terminology consistency — not validated for Amharic (https://www.researchgate.net/publication/358368108_Measuring_Terminology_Consistency_in_Translated_Corpora_Implementation_of_the_Herfindahl-Hirshman_Index)
WEB-27	JUST-NLP 2025 legal MT shared task uses BLEU/METEOR/COMET only, not for African languages (https://aclanthology.org/2025.justnlp-main.3.pdf)

Information Gaps

- Whether HHI or Semenov-Bojar terminology-consistency metrics can be implemented for Amharic with available resources
- Whether federal bureau translation units have internal scoring rubrics that could be adopted

Requires Expert Verification

- Federal bureau translator validation of which output categories matter most for binding legal documents

Remediation

Gap 1: No metric evaluates terminological consistency across clauses for proclamation-derived terms; lexical cohesion definition treats term repetition as a defect, opposite to legal-document requirements; GPT-4o judge unreliable for African languages.

Recommendation: Implement a term-consistency metric (Semenov-Bojar WMT 2022 [WEB-25] or HHI-based [WEB-26]) over a curated list of 30–80 deployment-critical Amharic terms. Replace GPT-4o-as-judge with native-speaker review by federal bureau translators for any high-stakes evaluation. Add structural-fidelity scoring (numbered-clause preservation) if

deployment evolves toward format-preserving translation.

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